MATH 448/548 (CPT S 430/530)

Numerical Analysis

Spring 2024

Tuesday, Thursday 1:30pm – 2:45pm Cleveland 30

INSTRUCTOR: V.S. Mano Manoranjan

http://math.wsu.edu/math/faculty/mano/

OFFICE HOURS: Tuesday, Thursday 12:00(noon) – 1:00pm Zoom

E-MAIL: mano@wsu.edu

Any email correspondence from you should have the word "Numerical" in the subject line. Otherwise, your correspondence will not go to the right mail folder for me to respond.

GRADER: Erik Holtrop (erik.holtrop@wsu.edu)

REFERENCE BOOK: *Numerical Analysis* (Third Edition)

David Kincaid and Ward Cheney

OUTLINE: The purpose of this course is to introduce numerical methods for solving mathematical problems that arise in science and engineering. The main topics that will be covered are,

- Numerical solutions of ordinary differential equations
- Direct and iterative methods for solving linear systems
- Numerical solutions for partial differential equations

GRADING:

The course grade will be based on

- Homework – 50 points *(Late homework will not be accepted or graded.)*
- In-class assignments – 50 points
- Midterm exam – 100 points
- Comprehensive final exam – 200 points

If you are enrolled in MATH 548 or CPT S 530 you must complete an individual project (to be arranged with the instructor). The project can earn a maximum of 50 points.

Grading Scale

A 93 – 100	**A-** 90 – 92	**B+** 87 – 89	**B** 83 – 86	**B-** 80 – 82	
C+ 77 – 79	**C** 73 – 76	**C-** 70 – 72	**D+** 67 – 69	**D** 63 – 66	**F** < 62

Attendance is imperative to be successful in this course.

Assignments should be submitted in class on the due date. Assignments left in my mailbox or found on my office floor will not be graded.

Students with Disabilities

Reasonable accommodations are available for students with a documented disability. If you have a disability and need accommodations to fully participate in this class, please either visit or call the Access Center (Washington Building 217; 509-335-3417) to schedule an appointment with an Access Advisor. All accommodations MUST be approved through the Access Center.

WSU Safety Measures

Classroom and campus safety are of paramount importance at Washington State University and are the shared responsibility of the entire campus population. WSU urges students to follow the "Alert, Assess, Act" protocol for all types of emergencies and the "Run, Hide, Fight" for an active shooter incident (URL below). Remain ALERT (through direct observation or emergency notification), ASSESS your specific situation and ACT in the most appropriate way to assure your own safety (and the safety of others if you are able). Please sign up for emergency alerts on your account at MyWSU. For more information on this subject, campus safety and related topics, please view the FBI's "Run, Hide, Fight video (https://oem.wsu.edu/emergency-procedures/active-shooter/) and visit the WSU safety portal (https://oem.wsu.edu/about-us/).

Academic Integrity

You are encouraged to work with your classmates on assignments. However, each student must turn in original work. No copying will be accepted. Students who violate WSU's Standards of Conduct for Students will receive an F as a final grade in this course, will not have the option to withdraw from the course and will be reported to the Office of the Dean of Students. Cheating is defined in the Standards for Student Conduct WAC 504-26-010(3). It is strongly suggested that you read and understand these definitions. https://conduct.wsu.edu.

Stay safe.